

SUKHDEV MARU

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SUMMARY

Electronics and Communication Engineering student at DAVV with a focus on Embedded Systems, VLSI, and digital design, and concurrently pursuing a BS in Data Science from Indian Institute of Technology Madras. Hands-on experience includes Embedded development on STM32F401RETx, FPGA design using Verilog on Boolean Board Spartan 7, and PCB design with hardware bring-up. Co-Founded a drone startup with supporting internships and applied projects.

INTERNSHIPS & TECHNICAL EXPERIENCE

FPGA Intern under C2S project – IET-DAVV

JUNE 2024 - AUG 2024

- Developed and simulated Verilog-based digital modules, including RTL design, testbench creation, and functional verification
- Implemented and validated a traffic light controller and FSM on Boolean Board Spartan 7

Engineering Intern – Indore Municipal Corporation (Electrical & Mechanical)

JUNE 2023 - AUG 2023

- Assisted in electrical systems analysis and field-level engineering operations
- Gained exposure to real-world infrastructure and multidisciplinary workflows

Drone Startup

SEPT 2025 - DEC 2025

- Worked on embedded control systems, electronics integration, and UAV system assembly
- Involved in component selection, and system-level hardware integration.

TECHNICAL SKILLS

- LANGUAGES : C/Embedded C, Python, Java, Assembly (ARM STM32, 8051/8085/AVR),
- MICROCONTROLLERS : STM32 NUCLEO F401RETx, Intel 8051, ATmega328P, ESP8266, Raspberry Pi
- VLSI & DIGITAL DESIGN : Verilog HDL, FPGA Boolean Board SPARTAN 7, Tanner EDA
- SPI, I2C, UART/USART, communication & networking
- PCB design using DIPTRACE, Linux/ Embedded Linux/ Windows, UAV systems

EDUCATION

B.Tech + M.Tech in Electronics spl. Embedded Systems, DAVV

2022-2027

Devi Ahilya Vishwavidyalaya Indore (M.P), GPA - 8.2

B.S Data Science IIT Madras

2023-2027

IIT Madras (T.N.), GPA - 6.0

Higher Secondary

2020

Nirmala Convent School Rajgarh M.P, Percentage - 80

PROJECTS

- **Precision Robotic Arm:** ARM Nucleo F401RE, Raspberry Pi 5, IMU Sensors, DC Motors, Python Programming
- **Sand Grain Size Mapping System (MoES):** Raspberry Pi 5, Embedded Linux GPS, camera module, IoT data upload
- **Traffic Light Controller:** Implemented FSM, simulated on FPGA Boolean Board SPARTAN 7, Verilog HDL,

CERTIFICATIONS & WORKSHOPS

- **PCB Design Workshop:** PCB layout, manual etching, hardware bring up, created own circuit
- **Drone WorkShop IIT Indore under SWAYAN scheme:** UAV electronics, flight control basics
- **NPTEL:** Drone Systems and Control, Soft Skills.
- **IIT Indore:** Swayan Drone Workshop.
- **IIT Madras (BS Data Science):** Diploma level in Programming and Data Science
- **IET DAVV:** C2S intern.
- **IIT Madras:** Automation & Inference, Machine Learning in Electronics, Magical Mathematics IIT Madras